

01 / HARDWARE ACTUATION

TB6612FNG Motor Driver



| High-Efficiency Dual MOSFET H-Bridge for Precision Motor Control.

ACTUATOR 26

FIELD NOTES / 2026

02 / HARDWARE OVERVIEW

Compact Powerhouse

Dual H-Bridge Module

The TB6612FNG is a dual-channel IC capable of independently driving two DC motors or one 4-wire bipolar stepper motor.

CURRENT CAPACITY**1.2A Continuous**

Drives 1.2A per channel continuously, with peak current capacity up to 3.2A for short acceleration bursts.

MOTOR VOLTAGE**VM: 2.5V to 13.5V**

Wide motor supply range ideal for LiPo batteries, AA packs, or 12V DC power sources.

MOSFET EFFICIENCY**Ultra-Low $R_{DS(ON)}$**

Internal MOSFET switches feature $\sim 0.5\Omega$ output resistance. Minimal heat loss and maximum battery runtime.

PROTECTION**Thermal Shutdown**

Built-in thermal cut-off circuit disables outputs if IC temperature exceeds safe threshold.



03 / BENCHMARK

Why Ditch L298N?

MOSFET vs BJT

Modern MOSFET technology outperforms legacy BJT drivers in every metric:

FEATURE	LEGACY L298N	TB6612FNG	ADVANTAGE
Internal Architecture	BJT Darlington Transistors	Power MOSFETs	95%+ Energy Efficiency
Voltage Drop	1.8V – 2.5V lost as heat	< 0.2V loss	Full battery voltage reaches motor
Heat Dissipation	Requires heavy aluminum heatsink	Runs cool without heatsink	Lightweight & compact design
Form Factor	Large bulky breakout board	Tiny 20mm x 20mm breakout	Ideal for small mobile robots



04 / HARDWARE PINOUT

Pin Reference

Logic vs Power

The module separates micro-controller logic inputs from heavy motor power rails:

LOGIC INPUTS (Microcontroller)	
STBY	Standby Control (Active HIGH)
PWMA	Motor A Speed (PWM 0-255)
AIN1 / AIN2	Motor A Direction Control
PWMB	Motor B Speed (PWM 0-255)
BIN1 / BIN2	Motor B Direction Control
VCC	Logic Power (2.7V – 5.5V)

POWER & OUTPUTS (Motors & Battery)	
VM	Motor Supply Voltage (2.5V – 13.5V)
A01 / A02	Motor A Output Terminal
B01 / B02	Motor B Output Terminal
GND	Common Ground (Logic & Motor)

05 / POWER ARCHITECTURE

Power Rails & Ground Isolation

Dual Power Supply Setup

Isolating micro-controller logic from motor voltage noise is essential for stability.

- ↘ **VCC (Logic Power)**
Connect to 3.3V or 5V from Arduino / ESP32 logic supply.
- ↘ **VM (Motor Power)**
Connect to battery pack (6V, 7.4V LiPo, 12V). Never power from Arduino 5V pin!
- ↘ **Common Ground**
GND of battery, motor driver, and microcontroller MUST be connected together.

CRITICAL RULE**Always Share Ground!**

Without a common GND reference between the Arduino and TB6612FNG, control signals will float, causing erratic motor jitter or failure to run.



H-Bridge Truth Table

Direction Matrix

Setting AIN1/AIN2 (or BIN1/BIN2) logic states controls motor direction and braking:

IN1	IN2	PWM	STBY	OPERATING MODE
HIGH	LOW	HIGH/PWM	HIGH	Forward Direction (CW)
LOW	HIGH	HIGH/PWM	HIGH	Reverse Direction (CCW)
HIGH	HIGH	ANY	HIGH	Short Brake (Active Brake)
LOW	LOW	ANY	HIGH	Stop / Coast (Freewheel)
LOW	X	X	LOW	Standby / Sleep Mode

07 / SAFETY & SLEEP

The STBY Pin Enable & Emergency Stop

MUST BE PULLED HIGH**STBY = HIGH to Run**

When STBY is HIGH, the internal H-Bridge FETs are enabled. You can connect STBY to an Arduino GPIO or directly to VCC.

COMMON TRAP**Why isn't my motor turning?**

If STBY is left disconnected, floating, or LOW, the chip will stay in standby mode. All motor outputs will remain completely dead!

LOW POWER SLEEP**STBY = LOW for Standby**

Pulling STBY LOW immediately disables all outputs and enters ultra-low power standby mode (<1μA current consumption).

08 / SPEED CONTROL

PWM Speed Control

PWMA & PWMB Pins

Modulating motor speed using Pulse-Width Modulation (0 to 255):

PWM = 0 (0% DUTY)

Zero Speed

Motor outputs receive 0V. The motor remains stationary.

PWM = 128 (50% DUTY)

Half Speed

Outputs pulse at 50% duty cycle, delivering approximately half effective voltage.

PWM = 255 (100% DUTY)

Full Speed

Outputs driven continuously HIGH. Motor runs at full battery voltage.

MINIMUM STALL THRESHOLD

Deadzone Voltage

DC motors require a minimum PWM threshold (typically 40–60) to overcome mechanical friction.

09 / CODE IMPLEMENTATION

Arduino Pin Setup & Initialization Code

Configuring GPIO control pins for Dual Motor Driver A & B:

```
// Pin Definitions for Motor Driver A & B

const int AIN1 = 4;    // Motor A Direction 1
const int AIN2 = 5;    // Motor A Direction 2
const int PWMA = 6;    // Motor A Speed (PWM Pin)

const int BIN1 = 7;    // Motor B Direction 1
const int BIN2 = 8;    // Motor B Direction 2
const int PWMB = 9;    // Motor B Speed (PWM Pin)

const int STBY = 3;    // Standby Pin

void setup() {
  pinMode(AIN1, OUTPUT); pinMode(AIN2, OUTPUT); pinMode(PWMA, OUTPUT);
  pinMode(BIN1, OUTPUT); pinMode(BIN2, OUTPUT); pinMode(PWMB, OUTPUT);
  pinMode(STBY, OUTPUT);

  digitalWrite(STBY, HIGH); // Enable driver out of standby
}
```

10 / CODE IMPLEMENTATION

Motor Control Helper Functions

Clean C++ functions for directional drive and short braking:

```
// Drive Motor A (dir: 1 = Forward, -1 = Reverse, 0 = Brake)
```

```
void driveMotorA(int speed, int dir) {  
    if (dir == 1) { // Forward  
        digitalWrite(AIN1, HIGH);  
        digitalWrite(AIN2, LOW);  
    } else if (dir == -1) { // Reverse  
        digitalWrite(AIN1, LOW);  
        digitalWrite(AIN2, HIGH);  
    } else { // Short Brake  
        digitalWrite(AIN1, HIGH);  
        digitalWrite(AIN2, HIGH);  
    }  
    analogWrite(PWMA, constrain(abs(speed), 0, 255));  
}
```

11 / CODE IMPLEMENTATION

Differential Drive Differential Robot Sketch

Complete autonomous motion loop for a 2-wheeled robot:

```
// Main Robot Motion Loop

void loop() {
  driveMotorA(200, 1); // Move Forward
  driveMotorB(200, 1);
  delay(2000);

  driveMotorA(150, -1); // Pivot Right
  driveMotorB(150, 1);
  delay(800);

  driveMotorA(0, 0); // Short Brake
  driveMotorB(0, 0);
  delay(1000);
}
```

12 / ENGINEERING RULES

4 Gold Rules of Motor Driver Wiring

RULE 1**Decoupling Capacitors**

Place a 10 μ F to 100 μ F electrolytic capacitor across VM and GND close to the board to absorb motor voltage spikes!

RULE 2**Current Limits**

Keep continuous motor current under 1.2A per channel. Stall current of cheap DC motors can exceed 2A!

RULE 3**Flyback Diodes**

TB6612FNG has internal flyback diodes. Avoid adding external diodes unless driving massive inductive loads.

RULE 4**Never Reverse Polarity**

Connecting VM and GND backwards will instantly burn the MOSFET driver IC! Always double-check battery wiring.



13 / READY TO DRIVE

Power. Control. Move.

You now have full mastery of the TB6612FNG Dual Motor Driver for your ACTUATOR 26 robotics builds.

TB6612FNG · H-BRIDGE · ACTUATOR 26

